Fermi–Amaldi-Based Local Mixing Function

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I. MAIN TEXT

Consider the exact exchange energy density (closed-shell):

$$e_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r}) = -\frac{1}{4}\int \frac{|\gamma(\mathbf{r};\mathbf{r}')|^2}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}', \tag{1}$$

where

$$\gamma(\mathbf{r},\mathbf{r}') = 2\sum_{i=1}^{N/2}\varphi_i^*(\mathbf{r})\varphi_i(\mathbf{r}') \tag{2}$$

is one-electron reduced density matrix [1, 2], and the Fermi–Amaldi [3, 4] energy density expressed via electron density, $\rho(\mathbf{r}) \geq 0$, and electron number, N,

$$e_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}) = -\frac{1}{2N}\int \frac{\rho(\mathbf{r})\rho(\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}'. \tag{3}$$

For one-orbital systems (one-electron and two-electron singlet), two quantities are equal for all values of $\mathbf{r}$ [5–7]:

$$e_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r}) = e_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}) \quad N \leq 2 \quad \text{for all } \mathbf{r}. \tag{4}$$

For systems with more than one molecular orbital, the spatial behavior of both quantities needs to be studied closer to establish a relationship between the exact and the Fermi–Amaldi exchnage energy densities. Thus, by the Cauchy–Schwarz inequality:

$$\rho(\mathbf{r})\rho(\mathbf{r}') \geq |\gamma(\mathbf{r},\mathbf{r}')|^2 \quad N > 2 \quad \text{for all } \mathbf{r}, \tag{5}$$

while the Fermi–Amaldi prefactor, $1/N$, decays faster than the constant prefactor, $1/2$, of the exact exchange hole. Therefore, there are some ranges of $|\mathbf{r}-\mathbf{r}'|$ dominated by one energy density and others dominated by another.

As $|\mathbf{r}| \to \infty$ while $|\mathbf{r}'|$ is fixed, both exchange energy densities are approaching the sme limit:

$$e_{\mathrm{x}}^{\mathrm{exact}}(|\mathbf{r}|\to\infty) = e_{\mathrm{x}}^{\mathrm{FA}}(|\mathbf{r}|\to\infty) \sim -\frac{\rho(\mathbf{r})}{2|\mathbf{r}|}. \tag{6}$$

The exchange hole, $\rho_{\mathrm{x}}(\mathbf{r},\mathbf{r}')$, is defined via

$$e_{\mathrm{x}}(\mathbf{r}) = \frac{1}{2}\int \frac{\rho(\mathbf{r})\rho_{\mathrm{x}}(\mathbf{r},\mathbf{r}')}{|\mathbf{r}-\mathbf{r}'|}d\mathbf{r}' \tag{7}$$

leading to expressions for the exact and the Fermi–Amaldi exchange holes, respectively:

$$\rho_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r},\mathbf{r}') = -\frac{1}{2}\frac{|\gamma(\mathbf{r},\mathbf{r}')|^2}{\rho(\mathbf{r})}, \tag{8}$$

$$\rho_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}') = -\frac{1}{N}\rho(\mathbf{r}'). \tag{9}$$

Both exchange holes are strictly non-positive, and both are integrating to -1 over all space.

As such, in the limit of infinitesimally small interelectronic distances ($|\mathbf{r}| \to |\mathbf{r}'|$) we have:

$$|\gamma(\mathbf{r},\mathbf{r})|^2 = \rho(\mathbf{r})^2, \tag{10}$$

consequently, the depths of the exchange holes are

$$\rho_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r},\mathbf{r}) = -\frac{\rho(\mathbf{r})}{2} \tag{11}$$

and

$$\rho_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}) = -\frac{\rho(\mathbf{r})}{N} \tag{12}$$

thus, for $N > 2$, the well of the exchange hole is dominated by the exact exchange hole.

Regarding the behavior far from the reference electron, the asymptotic decay of the Fermi–Amaldi hole follows that of the total electron density at large distances:

$$\rho_{\mathrm{x}}^{\mathrm{FA}}(\mathbf{r}') \sim -\frac{1}{N}\exp\left(-2\sqrt{2I_{\min}}|\mathbf{r}'|\right), \tag{13}$$

where $I_{\min}$ is the exact first ionization energy [2, 8]. For the exact exchange hole, the asymptotic behavior is given by:

$$\rho_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r},\mathbf{r}') \sim -\frac{2|\varphi_{\mathrm{HOMO}}(\mathbf{r})|^2}{\rho(\mathbf{r})}\exp\left(-2\sqrt{2I_{\min}}|\mathbf{r}'|\right), \tag{14}$$

given that of the molecular orbital [2, 8].

II. SCRAP

A local hybrid exchange functional is given by

$$E_{\mathrm{x}}^{\mathrm{LH}} = \int \left\{g(\mathbf{r})e_{\mathrm{x}}^{\mathrm{exact}}(\mathbf{r}) + [1-g(\mathbf{r})]e_{\mathrm{x}}^{\mathrm{SL}}(\mathbf{r})\right\}d\mathbf{r}, \tag{15}$$

where the integrand is a mixture of two exchange density energies weighed by a local mixing function (LMF), $g(\mathbf{r})$, which is position-dependent. The $e_{\mathrm{x}}^{\mathrm{SL}}$ and $e_{\mathrm{x}}^{\mathrm{exact}}$ quantities are a semi-local and the exact exchange energy densities, respectively.

A particular case of a one-electron limit, i.e., the asymptotic limit ($r \to \infty$) where the exact exchange must cancel out an excessive self-repulsion of the Coulomb potential, is also satisfied:

$$\lim_{r\to\infty}\frac{e_{\mathrm{x}}^{\mathrm{FA}}}{e_{\mathrm{x}}^{\mathrm{exact}}} = 1 \tag{16}$$

since both energy densities approach $-1/2r$ as $r \to \infty$.

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